DAY 03 OF 16 · 3D & 6DOF STREAMING

Your first 3D Python program

Import, extract, analyse, visualise, export: the four-step program from the book, run on a real file.

Environment | Import | Attributes | Analysis | Visualise & export

Book: Ch. 3 · Code: day03_first_3d_program/

BOOK CH. 3

Every 3D program follows the same four steps

01

Import

Read a file into a NumPy array (rows = points).

02

Extract

Slice columns: xyz, rgb, labels.

03

Analyse

Statistics, filters, per-class summaries.

04

Visualise & export

Open3D window, then .ply for other tools.

Our file has 7 columns: X;Y;Z;R;G;B;class, the same layout as the book’s ASCII examples.

BOOK CH. 3 · SETUP

A clean environment saves hours

Python 3.10–3.12

Open3D publishes wheels for these versions. Use a virtual environment per project.

01

Base libraries

NumPy for arrays, SciPy for k-d trees, Matplotlib for plots, Pillow for images.

02

3D libraries

Open3D for processing and viewing. Optional: laspy (LiDAR), PyVista, DracoPy.

03

STEP 2 · ATTRIBUTES

What each column means

Column	Type	Meaning	Typical use
X, Y, Z	float (m)	Position in the world frame	Geometry, distances, rendering
R, G, B	0–255	Colour seen by the sensor	Display, texture, segmentation
class	integer	Label: 0 ground, 1 wall, 2 roof…	Semantics, training, smart streaming
(intensity)	float	LiDAR return strength	Material, road markings
(normal)	3 floats	Surface direction	Lighting, meshing, features

STEP 3 · ANALYSIS

Filters turn questions into masks

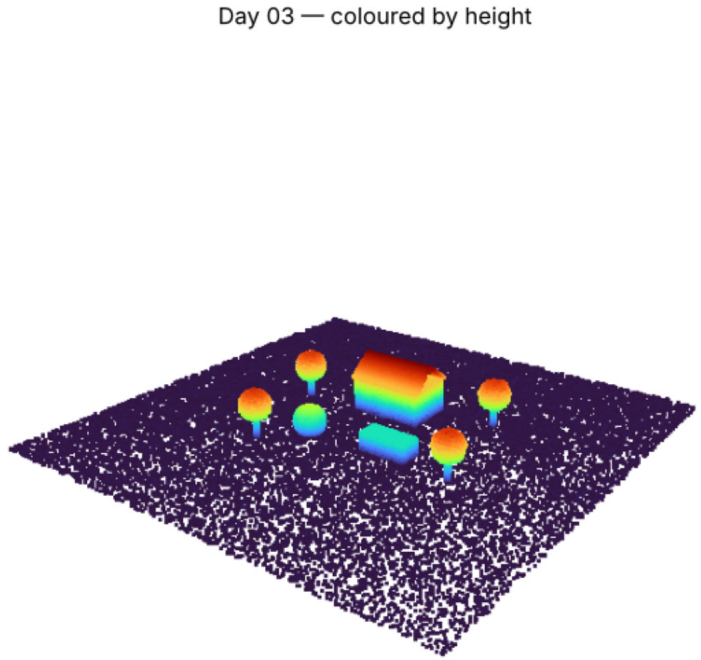


- above = xyz[:, 2] > 0.3
- keeps 59% of points: everything that is not ground
- Per class: count, mean height, extent
- One line of NumPy = one spatial question

outputs/no_ground.png: the mask z > 0.3 m removes the ground in one line of NumPy.

STEP 4 · VISUALISE & EXPORT

Colour by an attribute to see it



- Open3D window: rotate, zoom, pan
- write_ply → scene_full.ply, scene_no_ground.ply
- Same files open in MeshLab / CloudCompare

Export to .ply and open the file anywhere: MeshLab, CloudCompare, or 3dviewer.net in a browser.

HANDS-ON LAB · COPY, RUN, LOOK

Your first 3D program end to end

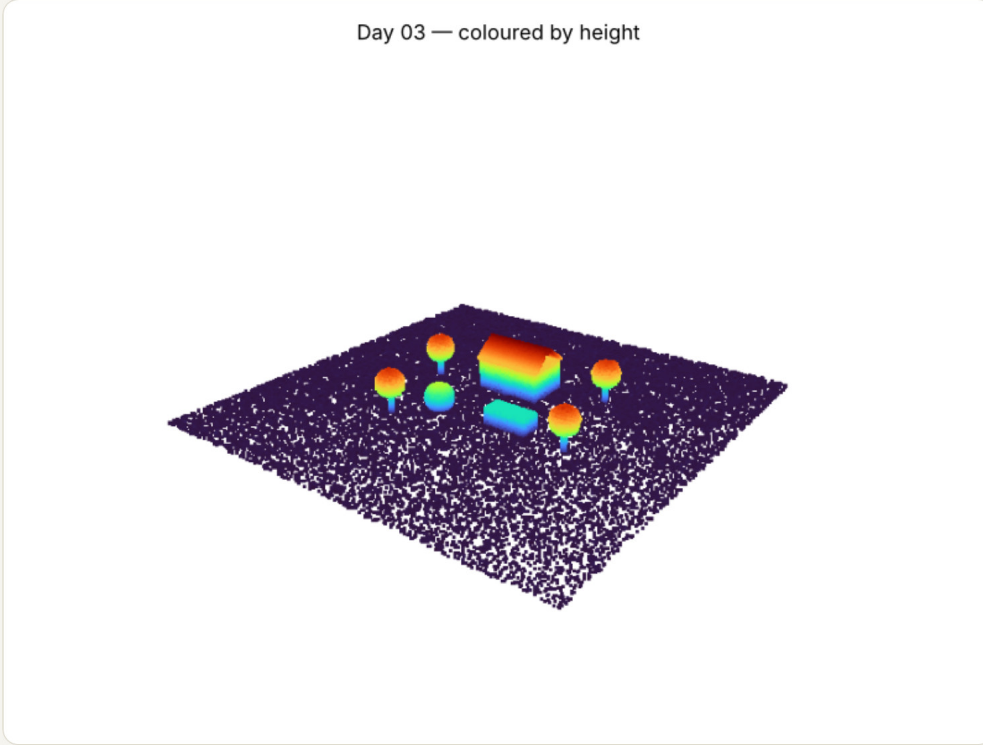
RUN IT

```
cd day03_first_3d_program
python first_program.py
# then: change the threshold,
# export only the roof (class 2)
```

→ Read the per-class table in the terminal.

→ Open outputs/*.ply in a viewer.

→ Try the three exercises in the README.



Three things to remember

1. A point cloud in Python is an (N, k) array: rows are points, columns are attributes.
2. Import → extract → analyse → visualise/export is the skeleton of every 3D script.
3. Masks on attributes answer spatial questions in one line.

BEFORE NEXT SESSION

Start Lab L2, questions 1 and 2. Next time we look at what all those bytes cost.